Next-event tokens

Token decoder

Linear output layer

Final LayerNorm

Causal self-attention block
within the next event

× 2

Token embeddings + h_i

h_i

Final LayerNorm

History decoder

6 ×

Feed forward
GELU, 384 to 1,536

LayerNorm 2

Causal multi-head attention
over events, temporal RoPE

LayerNorm 1

e_1 to e_i, one vector per event

Event encoder

Mean pool

Self-attention block
bidirectional, within each event

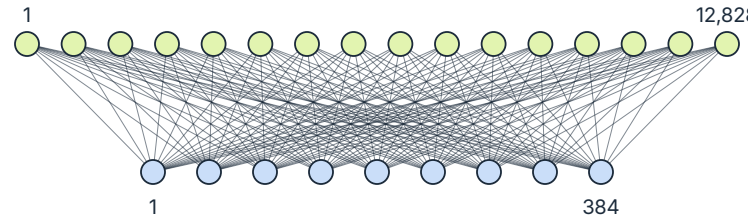
× 2

Token embedding layer

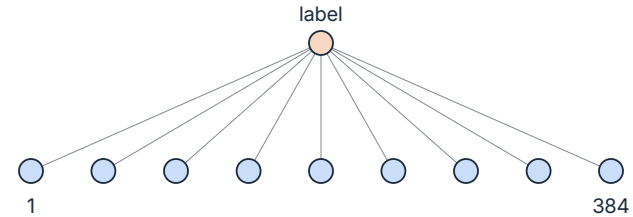
Tokenized events

EventGPT

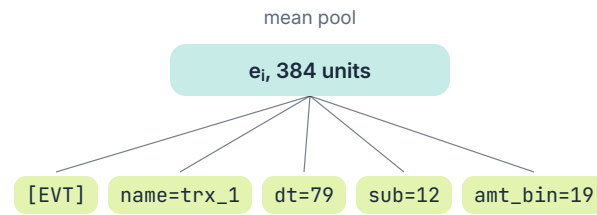
A client's events



The linear output layer maps 384 hidden units to 12,828 units, one per token in the vocabulary. Zero-shot scores read the target's tokens here.



For a supervised label, a logistic regression reads h_i at the as-of date, frozen: 384 hidden units to one probability.



The event encoder pools the tokens of each event into one vector, so the history decoder has one position per event and attends over events.